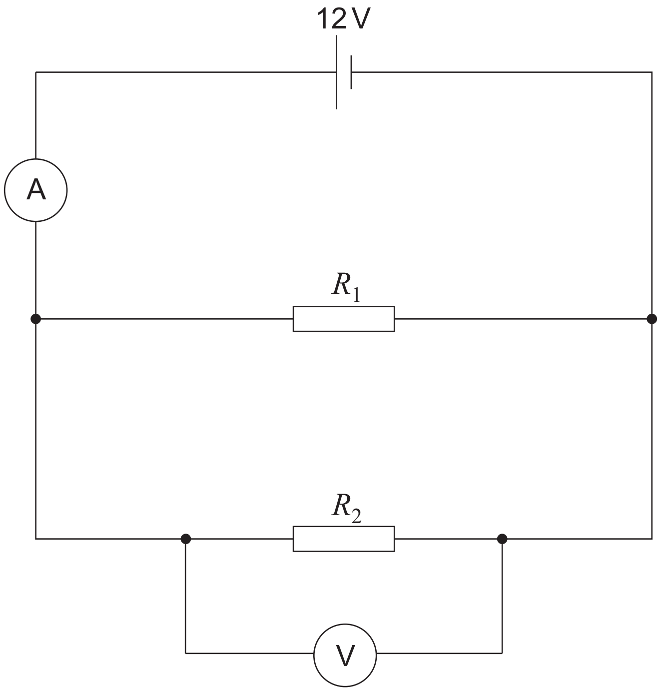
Examiner only

3. The circuit shown below contains two identical resistors connected in parallel. Each resistor has a resistance of 10 Ω.



(a) (i) Use the equation:

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

to calculate the total resistance of the circuit. [2]

total resistance = ... Ω

(ii) Use an equation from page 2 to calculate the current reading on the ammeter. [2]

current = ... A

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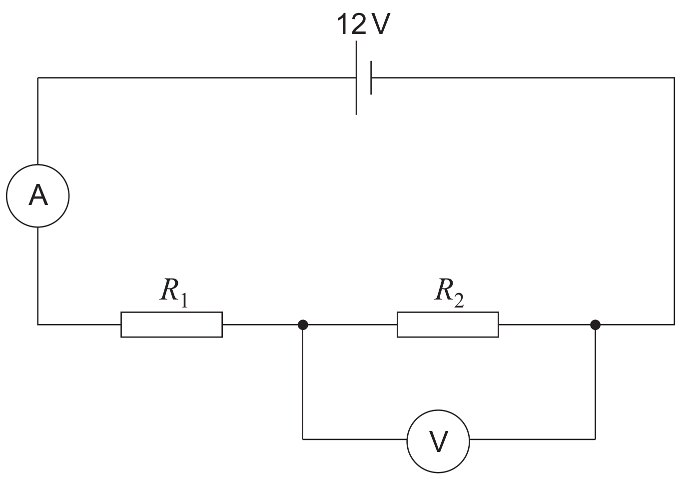
(iii) State the voltage across R_2. [1]

voltage = ... V

(iv) Calculate the current through R_2. [2]

current = ... A

(b) The same resistors are now connected in series.



State how the following values compare with the parallel circuit in part (a).

(i) The total resistance of the circuit. [1]

....

(ii) The current reading on the ammeter. [1]

....

(iii) I. The voltage across R_2. [1]

....

II. Give a reason for your answer. [1]

....

....

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